Vihan Patil

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Education

University of California, Santa Cruz
M.S., Computer Science and Engineering | GPA: 3.77/4.00 — *Sep 2024 – Mar 2026*
B.S., Computer Science — *Sep 2020 – Jun 2024*
Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Distributed Systems

Technical Skills

ML & AI: RAG/LLM pipelines, LLM evaluation, embeddings, vector search (FAISS, HNSW), knowledge graphs, reinforcement learning, PyTorch, TensorFlow, scikit-learn, OpenCV, pandas, NumPy
Languages: Python, SQL, TypeScript, JavaScript, Java, C, C++
Infrastructure & Tools: Docker, Kubernetes, AWS (EC2), GCP (Cloud Storage), Git, CI/CD, Linux, REST APIs, Node.js, Express.js, React

Experience

Graduate Researcher — AI Explainability & Accountability (AIEA) Lab, UCSC — *Sep 2024 – Mar 2026*
- Architected RAG pipelines with citation tracing and cross-model LLM verification, reducing hallucinations and improving answer relevance and interpretability.
- Prototyped a lightweight LLM evaluation harness scoring faithfulness, citation accuracy, and answer relevance over a curated knowledge base.
- Integrated neurosymbolic workflows combining knowledge-graph retrieval with transformer models to produce auditable, structured reasoning chains.

Software Engineering Intern — Vaara Drone — *Jun–Sep 2024 & Jun–Sep 2025*
- Designed an asynchronous multiprocessing pipeline for image ingestion and preprocessing, improving image-processing throughput by **75%** via optimized OpenCV and NDVI workflows.
- Simulated and evaluated reinforcement-learning landing policies for high-precision autonomous landing, validating policy stability and safety constraints.
- Engineered an edge-based drone mapping pipeline on Raspberry Pi with real-time object detection for low-latency area surveillance on constrained hardware.
- Created evaluation tooling around MAVLink command and telemetry streams to maintain reliable area mapping.

Teaching Assistant — Management of Technology (TIM 172), UC Santa Cruz — *Sep 2025 – Mar 2026*
- Coordinated grading workflows and rubrics for a 150+ student course; led project check-ins and office hours.

Projects

SwathKeeper — Autonomous Drone Object Avoidance + Survey Simulator (GitHub) — *In Progress*
- Building an autonomous crop-survey drone simulator on ArduPilot SITL, Gazebo, and ROS 2; demonstrated live reactive obstacle avoidance: AUTO→GUIDED→AUTO takeover, geofence-vetted 3D dodge, and mission resume.
- Enforced survey-coverage integrity with a ledger invariant that fails tests on silently skipped scan cells, validated by **94** automated tests in CI.

FinScreen — Financial Text-Signal Research Pipeline — *In Progress*
- Constructing an NLP pipeline over SEC EDGAR ingesting 10-K/10-Q/8-K filings (25 companies, 9 quarters) and extracting MD&A, risk-factor, and press-release text with point-in-time metadata.
- Produced a **6,400**-example weak-supervision labeled corpus (sentiment, red-flag taxonomy, guidance direction) via the Claude Batch API with boilerplate deduplication and look-ahead-bias-safe prompts.

Retrieval Algorithm Benchmarking (GitHub) — *Mar 2026*
- Benchmarked five ANN vector-index backends (FAISS Flat, HNSW, IVF-PQ, OPQ; hnswlib) via automated parameter sweeps over 50K-vector embedding datasets, measuring recall@k, p50/p95/p99 latency, QPS, memory, and index size.
- Mapped the recall/latency frontier to identify FAISS HNSW configurations sustaining perfect recall@10 at 54K QPS - a **3.3**× throughput gain over exact flat search at **3**× lower median latency with negligible index-size overhead.

HealthWise — Agentic RAG Health Coach (M.S. Capstone) — *Mar 2026*
- Built an agentic RAG chatbot delivering evidence-based, personalized nutrition and wellness coaching, grounding responses in a curated corpus of 60+ nutrition-research documents with citation-verified retrieval.

Blockchain Analytics — Industry-Client Project (UXly) — *Jun 2024*
- Developed a full-stack TypeScript/React + Node.js/Express app for real-time blockchain visualization and smart-contract analysis; scoped requirements with the client CEO and shipped to production on AWS EC2.